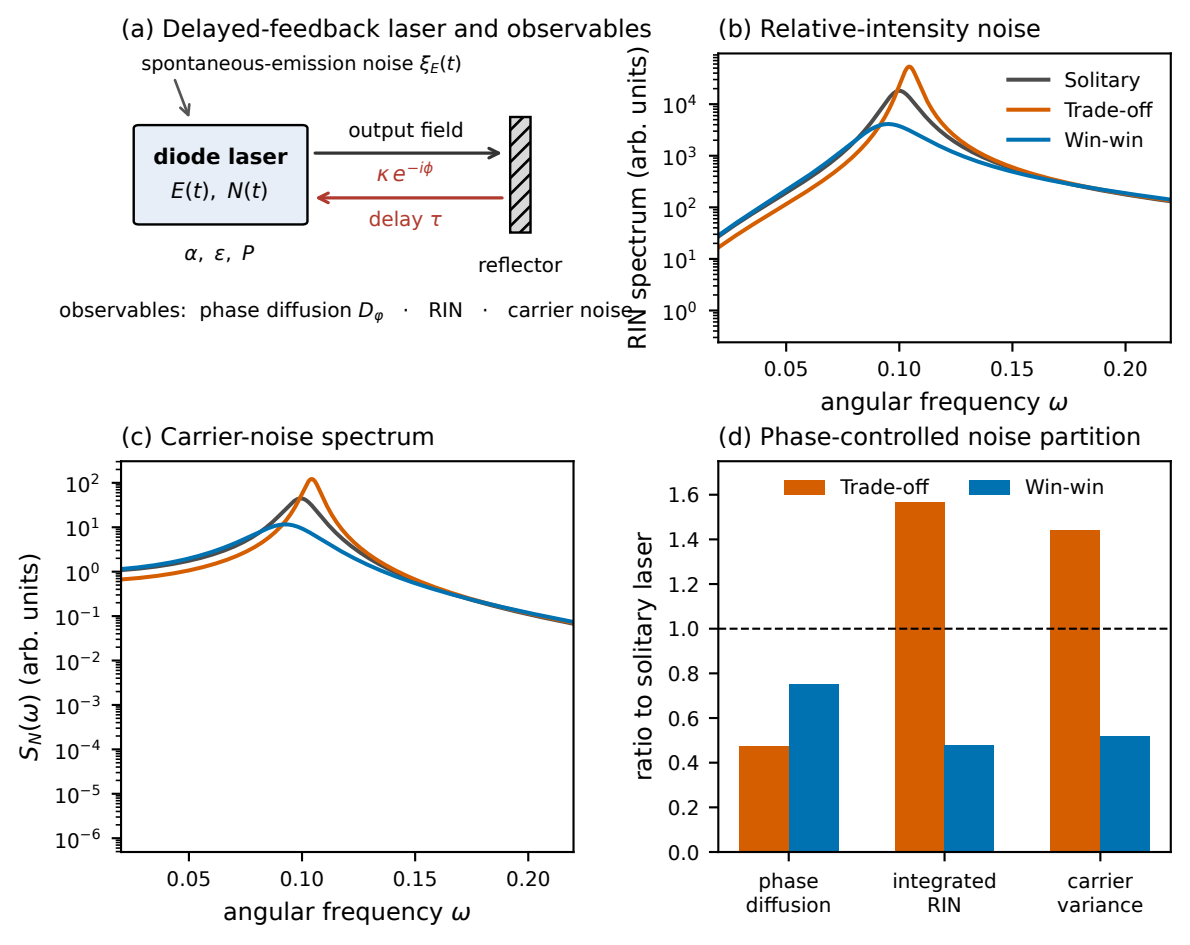
(a) Delayed-feedback laser and observables
spontaneous-emission noise $\xi_E(t)$
diode laser
$E(t),\ N(t)$
output field
$\kappa\, e^{-i\phi}$
delay τ
$\alpha,\ \varepsilon,\ P$
reflector
observables: phase diffusion D_φ · RIN · carrier noise
(b) Relative-intensity noise
RIN spectrum (arb. units)
Solitary
Trade-off
Win-win
angular frequency ω
(c) Carrier-noise spectrum
$S_N(\omega)$ (arb. units)
angular frequency ω
(d) Phase-controlled noise partition
Trade-off
Win-win
ratio to solitary laser
phase diffusion
integrated RIN
carrier variance